

Blockchain Experiment No: 7

AIM: Implement the embedding wallet (Metamask) and transaction using Solidity

Theory:

Blockchain is a decentralized and distributed digital ledger that records transactions securely across multiple systems. It ensures transparency, security, and immutability, meaning once data is recorded, it cannot be changed.

Ethereum is a popular blockchain platform that allows the execution of Smart Contracts. Smart contracts are self-executing programs written in Solidity that automatically perform actions when predefined conditions are met, eliminating the need for intermediaries.

MetaMask is a cryptocurrency wallet and browser extension used to interact with blockchain networks. It stores user accounts and private keys securely and allows users to send, receive, and sign transactions. It also acts as a bridge between Remix IDE and the blockchain network.

Testnets such as Sepolia and RSK Testnet are used for testing purposes. They provide free tokens, allowing developers to deploy and test smart contracts without using real cryptocurrency, reducing financial risk.

Remix IDE is a web-based development environment used to write, compile, deploy, and test smart contracts. It provides a user-friendly interface and supports direct integration with MetaMask for deploying contracts on blockchain networks.

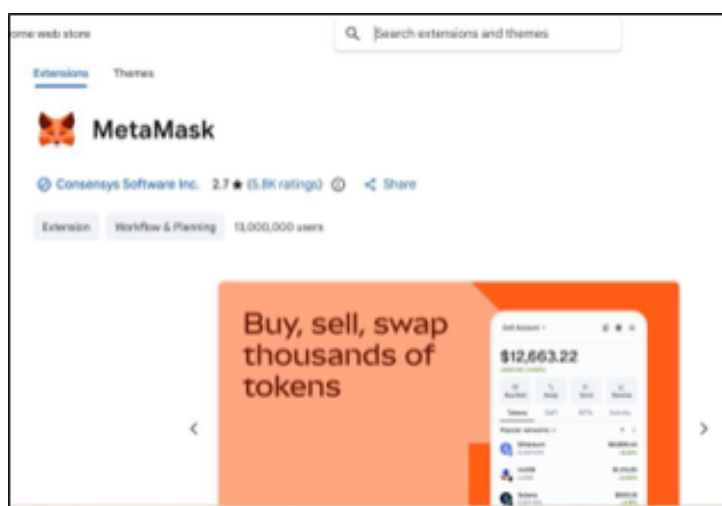
The smart contract deployment process includes writing the contract in Solidity, compiling it to remove errors, connecting MetaMask to Remix, and deploying the contract on a testnet by paying gas fees using test tokens.

A blockchain transaction refers to any interaction with the network, such as deploying a contract or calling its functions. Each transaction requires authentication through a digital signature and is permanently recorded on the blockchain. These transactions can be verified using blockchain explorers.

Implementation

Step 1: Install MetaMask

Install MetaMask extension and create a wallet





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MetaMask password

Losing this password means losing wallet access on all devices, **MetaMask can't reset it.**

Create new password

••••••••

👁

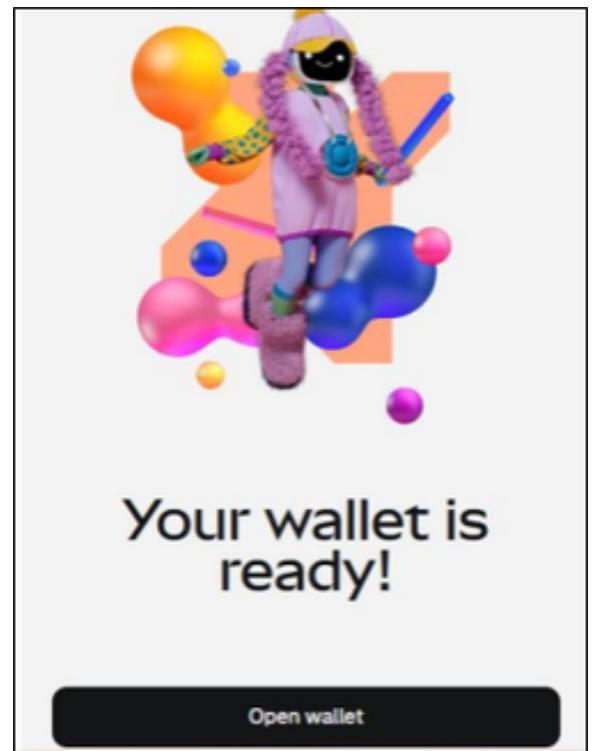
Must be at least 8 characters

Confirm password

••••••••

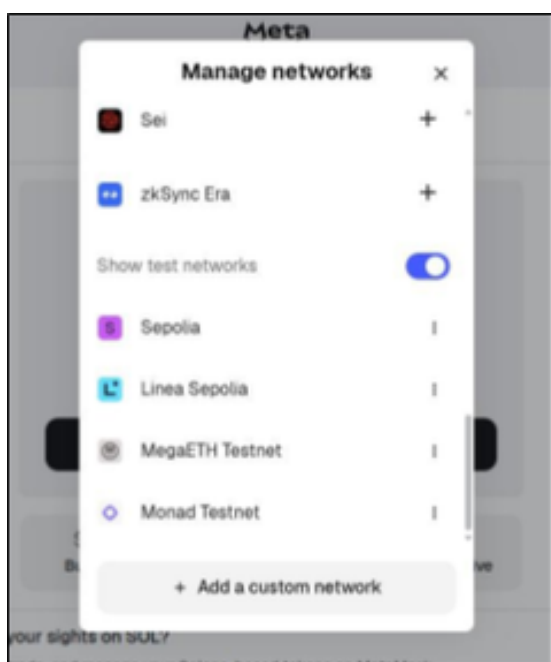
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☐ Get product updates, tips, and news including by email. We may use your interactions to improve what we share.



Step 2: Add Sepolia Testnet

Configure MetaMask to connect to Sepolia network.



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Edit network

×

Network name

Sepolia

Default RPC URL

Infura
sepolia.infura.io

Chain ID

11155111

Currency symbol

SepoliaETH

Save

Getting tokens

Your testnet tokens are being sent.

Requests for native tokens require the user to hold at least 1 LINK on Ethereum Mainnet. This requirement does not apply to requests for testnet LINK.

Ethereum-Sepolia
Drizzle 0.5.5 TH

0x9100c3b7a285192b4807a0b4e03aa03aa0000000000000000000000000000000000

Signature request

Review request details before you confirm.

Network: Sepolia

Request from: faucet.chainlink

Message: Welcome to Chainlink Faucet!
We require a signature in order to ensure you are the owner of the wallet requesting funds.

Wallet address: 0x9100c3b7a285192b4807a0b4e03aa0000000000000000000000000000000000

Cancel Confirm

Runtime Error

is now fixed

<

Add a custom network

×

Network name

RSK Testnet

According to our records, the network name may not correctly match this chain ID.
Suggested name: [Rootstock Testnet](#)

Default RPC URL

public-node.testnet.rsk.co

Chain ID

31

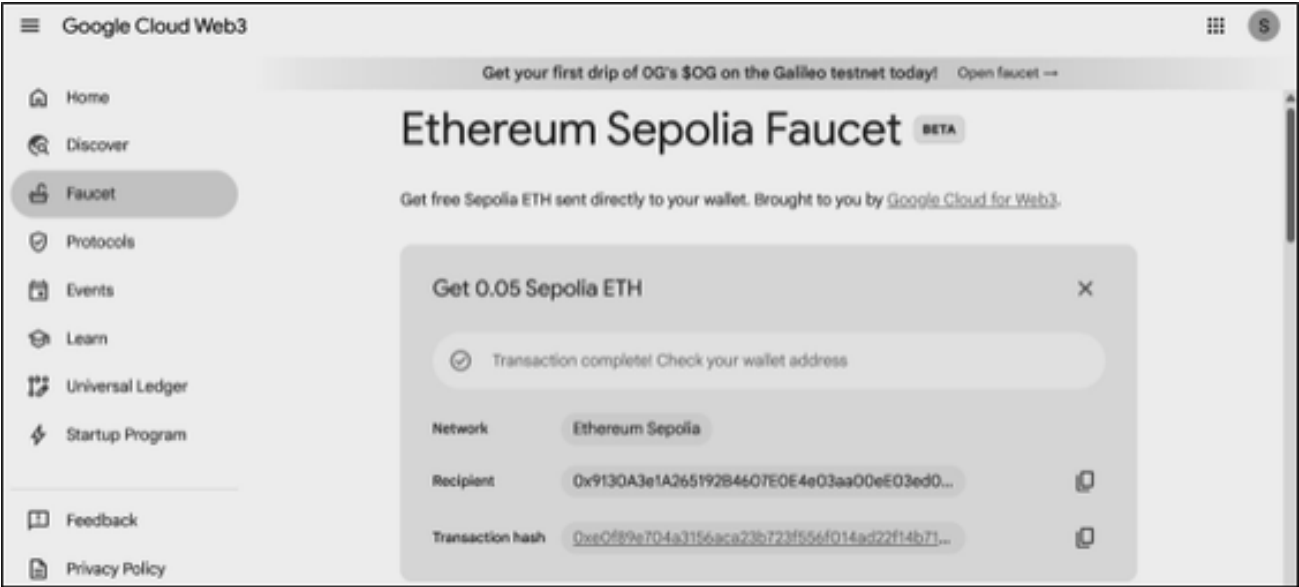
Currency symbol

us rsk

Save

Step 3: Fund Wallet

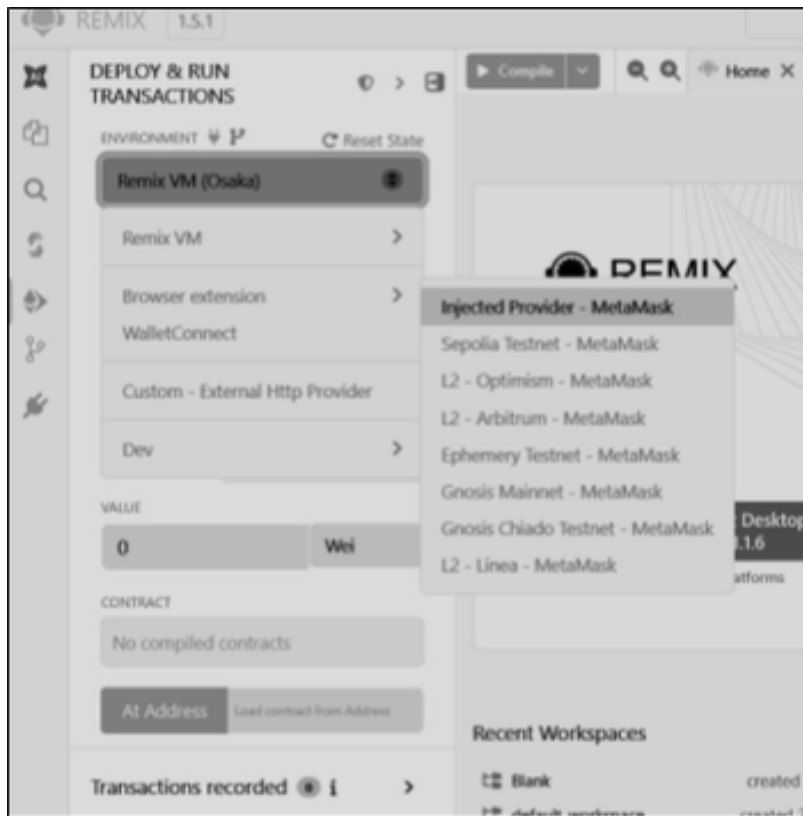
Use faucet to get test ETH

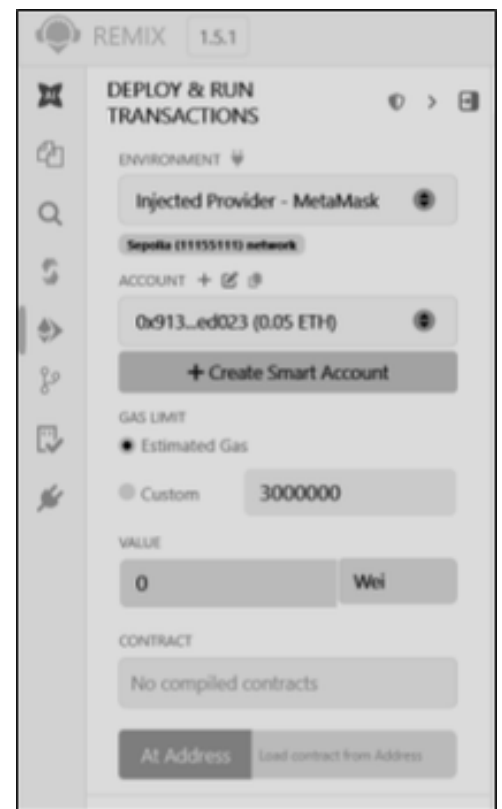


Step4: Connect the Sepolia Testnet / RSK Testnet to Remix IDE

1. In the Remix IDE, select the Environment as Injected Web3.

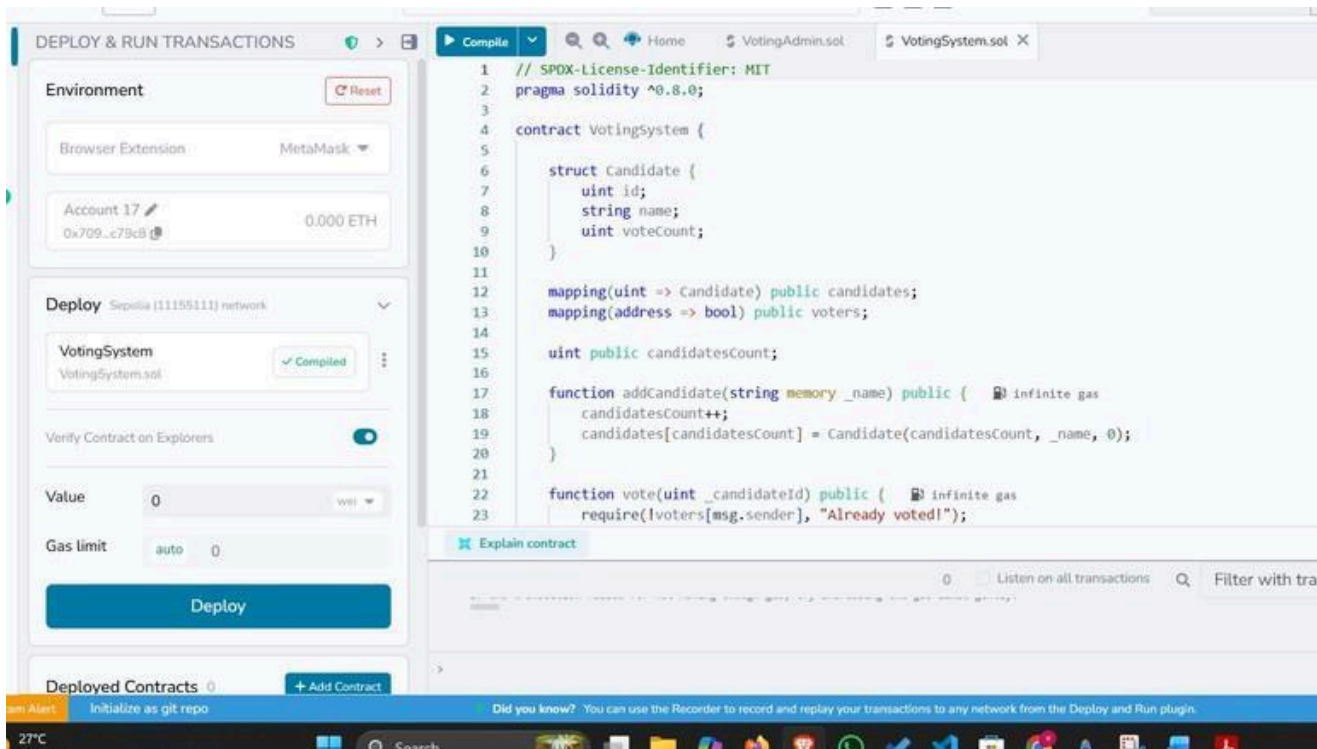
Note : Injected Web3 connects Remix with the active account in Metamask.





Compile the Contract

1. Open Remix
2. Paste your contract code
3. Go to Solidity Compiler (left panel)
4. Select version 0.8.x (match ^0.8.0)
5. Click Compile Votingsystem.sol



Connect MetaMask (Injected Provider)

1. Go to Deploy & Run Transactions tab
2. In Environment → select Injected Provider - MetaMask
3. MetaMask popup will open → click Connect Make sure:
 - Network = Sepolia Test Network
 - You have 0.05 Sepolia ETH

Environment

Reset

Browser Extensi... MetaMask ▼

Account 19 
0x862...c1199  0.000 ETH

Deploy Sepolia (11155111) network ▼

VotingSystem
VotingSystem.sol

✓ Compiled ⋮

Verify Contract on Explorers ☒

Value 0 wei ▼

Gas limit custom 3000000

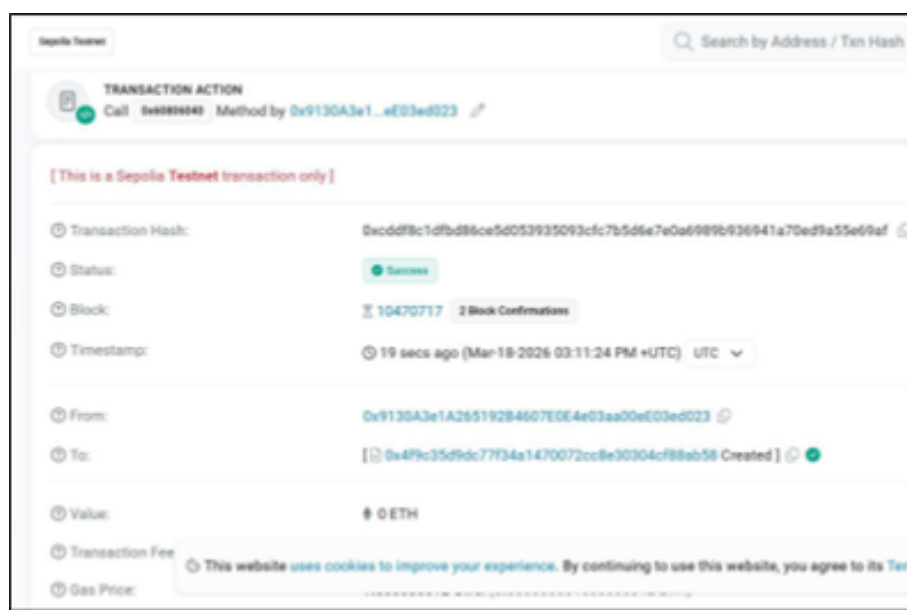
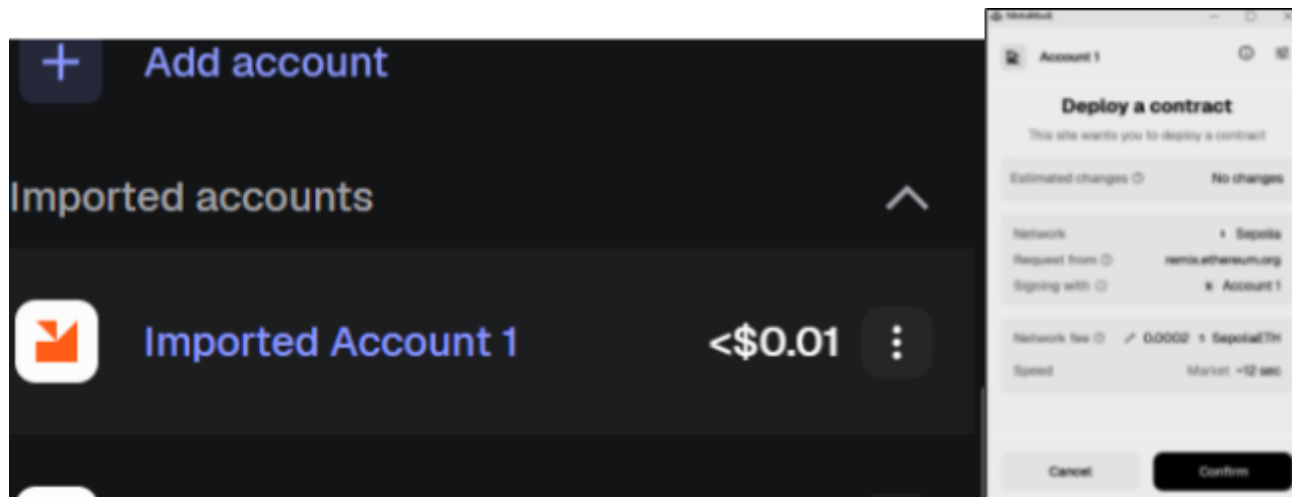
Deploy

Deploy Contract

Your constructor needs a Voting address Steps:

1. Copy your second MetaMask account address (voting)
 - MetaMask → Account 2 → Copy address
2. Paste that address in constructor field
3. Click Deploy
4. MetaMask will pop up → Confirm transaction Wait a few seconds...

After success → your contract appears under Deployed Contracts



Contract deployment

Status

Confirmed

View on block explorer

Copy transaction ID

From

Ox9f30A...ed023

New contract

To

Transaction

Nonce

1

Amount

-0 SepoliaETH

Gas Limit (Units)

826833

Gas Used (Units)

82149

Base fee (GWEI)

0.000000012

Priority fee (GWEI)

1.5

Total gas fee

0.005232 SepoliaETH

Max fee per gas

0.000000002 SepoliaETH

Total

0.00123172 SepoliaETH

DEPLOY & RUN TRANSACTIONS

0x9f30A...ed023 @04753655298

Create Smart Account

GAS LIMIT

Estimated Gas

Custom 3000000

VALUE

0 Wei

CONTRACT

PharmaSupply - PharmaSupply.sol

view source code

Verify Contract on Etherscan

Deploy & verify address: pharmacy

At Address Load contract from Address

Transactions recorded 1

Deployed Contracts

PHARMASUPPLY AT 0x812...801

Welcome to Remix 1.5.1

Your files are stored in IndexedDB, 398.24 KB /

You can use this terminal to:

- Check transaction details and start debugging.
- Execute JavaScript scripts:
 - Input a script directly in the command line
 - Select a Javascript file in the file explorer
 - Right-click on a Javascript file in the file explorer

The following libraries are accessible:

- ethers.js

Type the library name to see available commands or create of PharmaSupply pending....

view on Etherscan view on Blockscout

[block:18478717 txIndex:46] from: 0x9f30A...ed023

[verification] Contract deployed. Checking expl

Etherscan verification skipped: API key not pro

Please input the API key in Remix Settings - in

[Blockscout] Verification submitted. Awaiting c

[Morescan] Verification submitted. Awaiting c

[Sourceify] Verification Error: Contract notDef

[Sourceify] Please open the "Contract Verificat

[Blockscout] Verification Successful! View Co

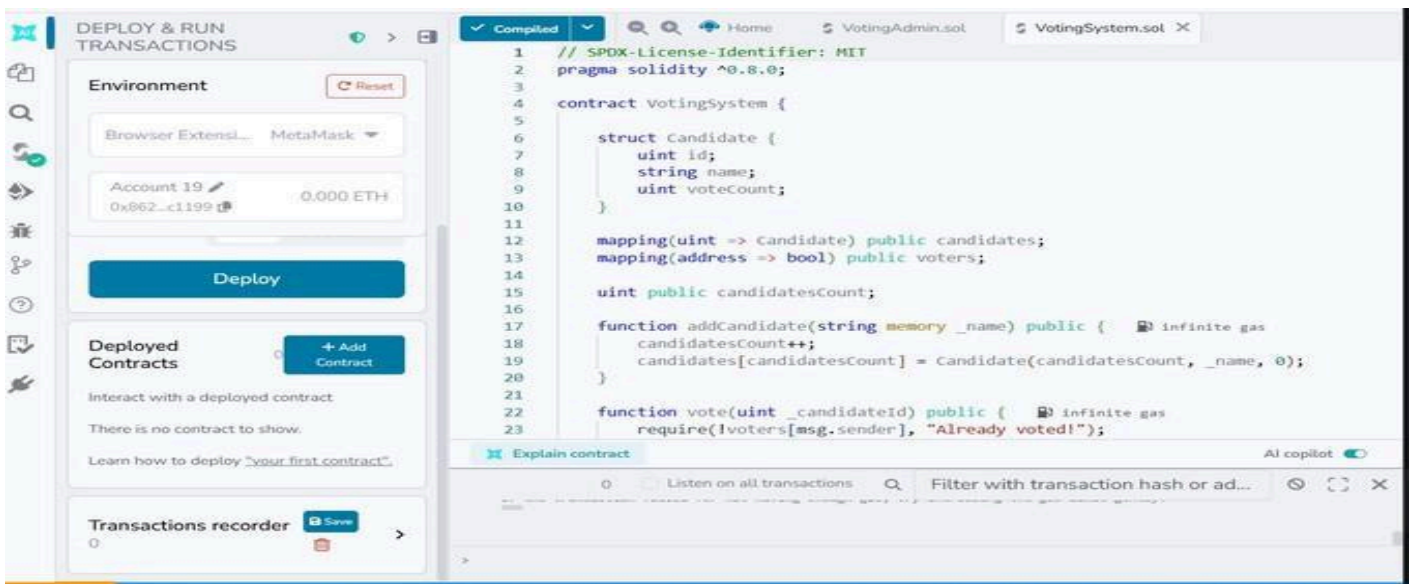
[Morescan] Verification Successful! View Co



Interact with Contract

Step 1: Add Medicine (Manufacturer only)

- Make sure MetaMask is on Account 1 (manufacturer)
- In Remix:
 - Enter:
 - `_candidate`: "Alice"
- Click addCandidate Confirm in MetaMask



Step 2: Calculate Vote

- Still on Account 1
- Click calculateVote
- Confirm transaction

[view on etherscan](#) [view on blocksout](#)

[block:10470765 txIndex:24] from: 0x913...e0023 to: PharmaSupply.shipMedicine() 0xbdc2...b6cf8 val: data: 0x39f...32075 logs: 0 hash: 0x9cd...6ffe2

status

1 Transaction mined and execution succeed

transaction hash

0x2c41b01fa5f9c8e66c4a8a4f55b3d2d9340a5f8b9f2ec8c1caf3337e880

block hash

0x9c092762208d4c4297e5f1674d25516e04d0f0030341892096c5d699ca16ffe2

block number

10470765

from

0x9130A3e1A20519204687E0E4e03ae06c01ed823

to

PharmaSupply.shipMedicine() 0xbdc2ef31a07e38cA204f70f091a0C8D2a5efb1cF8

transaction cost

45570 gas

decoded input

()

decoded output

-

logs

[]

raw logs

[]

Account 1

Transaction request

Estimated changes

No changes

Network

Sepolia

Request from

remix.ethereum.org

Interacting with

0xbdc2ef_b6cf8

Network fee

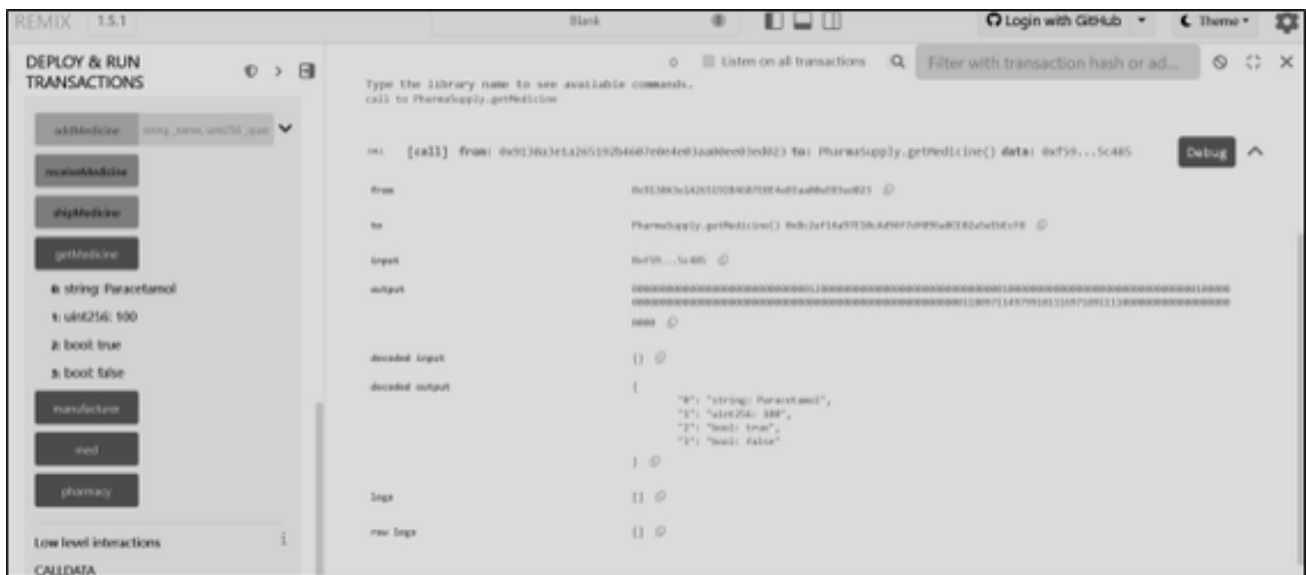
0.0001 SepoliaETH

Speed

Market ~12 sec

Cancel

Confirm



Conclusion

The experiment was successfully carried out by integrating the MetaMask wallet with Remix IDE and performing transactions on the Sepolia test network. The smart contract was compiled, deployed, and interacted with using MetaMask, which enabled secure transaction signing and real-time confirmation on the blockchain.

The use of a test network allowed safe execution without real financial risk, while also providing practical exposure to gas fees, transaction validation, and blockchain explorers. This experiment helped in understanding wallet integration, smart contract deployment, and the overall working of decentralized applications.

Thus, the objective of implementing MetaMask and performing blockchain transactions was successfully achieved.